

30.6 Summary and Notes

30.6.1 Summary

Idea.

1. A penalized criterion (Definition 30.1.1) has a minimizer when it is coercive; with a convex penalty all minimizers share their fit and penalty, and strict convexity or full rank gives uniqueness (Theorem 30.1.1).
2. For convex penalties, $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \in \lambda \partial P(\hat{\boldsymbol{\beta}})$ characterizes minimizers (Lemma 30.1.2, Lemma 30.1.3), and penalized estimates are posterior modes (Proposition 30.1.4).
3. SCAD and MCP are sparse, continuous and nearly unbiased; MCP is firm thresholding (Proposition 30.1.5).
4. The elastic net is unique, a lasso on augmented data, and groups correlated regressors (Proposition 30.2.1, Proposition 30.2.3).
5. The group lasso selects whole groups by block soft thresholding (Proposition 30.3.1) and, orthonormalized, ignores the coding of factors (Proposition 30.3.2).
6. Componentwise boosting converges geometrically to least squares (Theorem 30.4.1); early stopping is a spectral filter like ridge (Proposition 30.4.2), and the operator trace understates its degrees of freedom.
7. Forward stagewise tracks the lasso path (Proposition 30.5.1); its limit is the lasso when paths are monotone and the monotone lasso otherwise (Proposition 30.5.2).

30.6.2 Notes and sources

Coverage. Fahrmeir, Kneib, Lang and Marx (2021, §4.2–4.3) treat penalties and componentwise boosting in applied terms; the theory and proofs are added here. Hastie, Tibshirani and Friedman (2009, chapters 3, 10 and 16) cover the same ground from the machine-learning side.

Penalties. Ridge regression is due to Hoerl and Kennard (1970), the lasso to Tibshirani (1996) and bridge penalties to Frank and Friedman (1993); uniqueness of the lasso in general position is from Tibshirani (2013). Boyd and Vandenberghe (2004) cover subgradients, and Park and Casella (2008) develop the Bayesian lasso. SCAD and the oracle property are from Fan and Li (2001), MCP from Zhang (2010).

Elastic net and groups. The elastic net, its grouping effect and rescaling are from Zou and Hastie (2005); our grouping bound adds the residual-norm term. Pathwise coordinate descent is from Friedman, Hastie, Höfling and Tibshirani (2007) and Friedman, Hastie and Tibshirani (2010). The group lasso is due to Yuan and Lin (2006), its orthonormalized form is discussed by Simon and Tibshirani (2012), and the sparse group lasso is from Simon, Friedman, Hastie and Tibshirani (2013).

Boosting. AdaBoost is due to Freund and Schapire (1997); Friedman, Hastie and Tibshirani (2000) and Friedman (2001) gave the statistical and gradient view, and matching pursuit is from Mallat and Zhang (1993). L_2 boosting and the boosting operator are from Bühlmann and Yu (2003), componentwise boosting in high dimensions from Bühlmann (2006), and a survey is Bühlmann and Hothorn (2007). Our proof of Theorem 30.4.1 is the standard argument for greedy

coordinate descent on a quadratic. Efron, Hastie, Johnstone and Tibshirani (2004) proved that FS_0 is the lasso path under their positive cone condition, and Hastie, Taylor, Tibshirani and Walther (2007) described FS_0 in general as the monotone lasso; Proposition 30.5.1 is a special case proved directly.

30.6.3 References

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